

# RF and Microwave Engineering Numerical Companion

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · Chapter 4, all 8 PYQ waveguide and junction problems, 2069–2082 BS

## 4. RF/Microwave Components and Devices: Numerical Problems

8 solved PYQ problems · 46 marks across 8 papers · 2 guide-parameter sets, 1 cut-off family, 2 mode rankings, 3 junction networks

### READ THIS FIRST

Chapter 4 numerals come in two kinds, and neither needs more than a calculator's square root.

- **Waveguide arithmetic** (Problems 1 to 5). Everything hangs off one number, the **factor**  $F = \sqrt{1 - (f_c/f)^2}$ . Compute it once, write it in a box, and every other quantity is a multiplication or a division by it:

$$\lambda_g = \frac{\lambda}{F}, \quad v_p = \frac{c}{F}, \quad v_g = cF, \quad Z_{TE} = \frac{\eta_0}{F}, \quad Z_{TM} = \eta_0 F.$$

**Always start by naming the dominant mode and checking  $f > f_c$** ; a guide below cut-off has no  $\lambda_g$  or  $v_p$  to find.

- **Junction networks** (Problems 6 to 8). Treat each tee as its §3.6 matrix and **follow one unit wave through**: out of one tee, into the next, out again. A matched termination deletes a row and column; a connection hands a leaving wave to the other tee as an arriving one. Column power sums are the self-check.
- Constants used by the papers and here:  $c = 3 \times 10^8$  m/s,  $\eta_0 = 120\pi = 376.99 \Omega$ . Every answer is asserted in `src/wg.py`.

### Problem 1: Full parameter set of an air-filled rectangular guide 6

80 Ba · width 3 cm, 6 GHz, dominant mode

#### QUESTION AS PRINTED 80 Ba [6+4]

For an air-filled rectangular waveguide with a width of 3 cm and a desired frequency of operation of 6 GHz (for dominant mode), determine cut-off frequency, cut-off wavelength, group velocity, phase velocity, propagation wavelength in the waveguide and the characteristic impedance. Explain the four basic parameters used to describe the performance of a directional coupler.

Given

- Air filling:  $v = c = 3 \times 10^8$  m/s,  $\eta = \eta_0 = 120\pi \Omega$ .
- Width (broad wall)  $a = 3$  cm = 0.03 m;  $f = 6$  GHz.
- Dominant mode of a rectangular guide:  $\text{TE}_{10}$  ( $m = 1, n = 0$ ), so  $b$  is not needed.

#### Step 1: cut-off frequency and wavelength of $\text{TE}_{10}$

- $f_c = \frac{c}{2} \sqrt{\left(\frac{1}{a}\right)^2 + 0} = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 0.03} = \boxed{5 \text{ GHz}}$ .
- $\lambda_c = 2a = \boxed{6 \text{ cm}}$  (check:  $c/f_c = 3 \times 10^8 / 5 \times 10^9 = 0.06$  m).
- $f = 6 \text{ GHz} > f_c = 5 \text{ GHz} \Rightarrow$  **the guide propagates**, so the remaining quantities exist.

#### Step 2: the factor, computed once

- $f_c/f = 5/6 = 0.83333$ ;  $(f_c/f)^2 = 0.69444$ ;  $1 - 0.69444 = 0.30556$ .
- $F = \sqrt{0.30556} = \boxed{0.55277}$  (exactly  $\sqrt{11/6}$ ).
- Free-space wavelength  $\lambda = c/f = 3 \times 10^8 / 6 \times 10^9 = 5$  cm.

**Step 3: every other quantity from  $F$** 

| Quantity         | Formula                  | Substitution                   | Value                                     |
|------------------|--------------------------|--------------------------------|-------------------------------------------|
| Group velocity   | $v_g = cF$               | $3 \times 10^8 \times 0.55277$ | <b><math>1.658 \times 10^8</math> m/s</b> |
| Phase velocity   | $v_p = c/F$              | $3 \times 10^8 / 0.55277$      | <b><math>5.427 \times 10^8</math> m/s</b> |
| Guide wavelength | $\lambda_g = \lambda/F$  | $5/0.55277$                    | <b>9.045 cm</b>                           |
| Impedance (TE)   | $Z_{TE} = \eta_0/F$      | $376.99/0.55277$               | <b>682.0 <math>\Omega</math></b>          |
| Phase constant   | $\beta = 2\pi/\lambda_g$ | $2\pi/0.090453$                | 69.46 rad/m                               |

- **Checks:**  $v_p v_g = 5.427 \times 10^8 \times 1.658 \times 10^8 = 9.0 \times 10^{16} = c^2 \checkmark$ ; and  $\frac{1}{\lambda^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}$ :  $400 = 122.2 + 277.8 \text{ m}^{-2} \checkmark$ .
- “Characteristic impedance” of a TE guide is its wave impedance  $Z_{TE}$ ; it is above  $\eta_0$ , as it must be for TE.
- **Rider (4 marks):** coupling, directivity, isolation, insertion loss, with  $I = C + D$ . See theory §4.6.

**Problem 2: Same parameters, with suitably assumed dimensions 6**

77 Ch · the choice of dimensions is part of the marks

**QUESTION AS PRINTED** 77 Ch [6]

For a rectangular waveguide, with suitably assumed breadth, width and frequency, (for dominant mode), determine cut-off frequency, phase velocity, propagation wavelength in the waveguide and the characteristic impedance.

**Step 0: choose the data, and justify it**

- Take the standard X-band guide **WR-90**:  $a = 2.286 \text{ cm}$ ,  $b = 1.016 \text{ cm}$ , air filled, at  $f = 10 \text{ GHz}$ .
- Why it is a good choice: it runs in **single-mode**  $\text{TE}_{10}$  at 10 GHz. The next modes are cut off:
  - $\text{TE}_{20}$ :  $f_c = c/a = 3 \times 10^8 / 0.02286 = 13.12 \text{ GHz} > 10 \text{ GHz}$ .
  - $\text{TE}_{01}$ :  $f_c = c/2b = 3 \times 10^8 / 0.02032 = 14.76 \text{ GHz} > 10 \text{ GHz}$ .
- Any other data is acceptable if it is stated and above cut-off; the method does not change.

**Step 1: cut-off**

- $f_c = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 0.02286} = \boxed{6.562 \text{ GHz}}$ ,  $\lambda_c = 2a = 4.572 \text{ cm}$ .  $10 > 6.562 \checkmark$  propagates.

**Step 2: the factor**

- $f_c/f = 0.65617$ ; squared 0.43056;  $1 - 0.43056 = 0.56944$ ;  $F = \sqrt{0.56944} = \boxed{0.75461}$ .
- $\lambda = c/f = 3 \text{ cm}$ .

**Step 3: the asked quantities**

| Quantity               | Formula                 | Substitution                   | Value                                     |
|------------------------|-------------------------|--------------------------------|-------------------------------------------|
| Phase velocity         | $v_p = c/F$             | $3 \times 10^8 / 0.75461$      | <b><math>3.976 \times 10^8</math> m/s</b> |
| Guide wavelength       | $\lambda_g = \lambda/F$ | $3/0.75461$                    | <b>3.976 cm</b>                           |
| Impedance (TE)         | $Z_{TE} = \eta_0/F$     | $376.99/0.75461$               | <b>499.6 <math>\Omega</math></b>          |
| Group velocity (extra) | $v_g = cF$              | $3 \times 10^8 \times 0.75461$ | $2.264 \times 10^8 \text{ m/s}$           |

- Check:  $1/\lambda^2 = 1111.1$  and  $1/\lambda_g^2 + 1/\lambda_c^2 = 632.7 + 478.4 = 1111.1 \text{ m}^{-2} \checkmark$ .

**Problem 3: Does the guide pass the signal? Cut-off checks 3 2**

70 Bh, 80 Bh · one method, two instances; 70 Bh worked in full

**QUESTION AS PRINTED** 70 Bh [7+3]

Define expressions for various field components of a rectangular waveguide in TE mode. Show that a 1 GHz signal cannot propagate in  $\text{TE}_{10}$  made in a rectangular waveguide with a wall separation of 5 cm.

## QUESTION AS PRINTED 80 Bh [1+1+2]

What is dominant mode in a waveguide? What are the dominant modes for rectangular and circular waveguides? Does a rectangular waveguide of width 2.254 cm support the propagation of a signal having frequency 6 GHz?

## Method

1. Name the mode (TE<sub>10</sub>, the dominant one) and read the broad-wall width  $a$ .
2.  $f_c = c/2a$ .
3. Compare:  $f > f_c$  propagates;  $f < f_c$  is evanescent.
4. For “show that”, add the attenuation:  $\gamma = \alpha = \sqrt{k_c^2 - k^2}$  is real.

70 Bh worked:  $a = 5$  cm,  $f = 1$  GHz

- $f_c = \frac{3 \times 10^8}{2 \times 0.05} = 3 \times 10^9$  Hz = 3 GHz.
- $f = 1$  GHz  $< f_c = 3$  GHz  $\Rightarrow$  below cut-off.
- Confirm with  $\gamma$ :
  - $k_c = \pi/a = \pi/0.05 = 62.832$  rad/m,  $k_c^2 = 3947.8$ .
  - $k = 2\pi f/c = 2\pi \times 10^9/3 \times 10^8 = 20.944$  rad/m,  $k^2 = 438.6$ .
  - $\gamma^2 = k_c^2 - k^2 = 3947.8 - 438.6 = 3509.2 > 0 \Rightarrow \gamma$  is real:  $\gamma = \alpha = \sqrt{3509.2} = 59.24$  Np/m,  $\beta = 0$ .
- The field decays as  $e^{-59.24z}$ ; to  $1/e$  in  $1/59.24 = 1.69$  cm, i.e.  $20 \log_{10} e \times 59.24 = 514.5$  dB/m. **No wave travels: the 1 GHz signal cannot propagate.**

## Both instances

| Paper | Given                 | $f_c = c/2a$                       | Compare    | Answer                                         |
|-------|-----------------------|------------------------------------|------------|------------------------------------------------|
| 70 Bh | $a = 5$ cm, 1 GHz     | 3.00 GHz                           | $1 < 3$    | <b>cannot propagate</b> ; $\alpha = 59.2$ Np/m |
| 80 Bh | $a = 2.254$ cm, 6 GHz | $3 \times 10^8/0.04508 = 6.65$ GHz | $6 < 6.65$ | <b>not supported</b>                           |

- **80 Bh theory part:** dominant mode = lowest cut-off mode; rectangular TE<sub>10</sub>, circular TE<sub>11</sub> (§4.2).
- The handwritten local example (Waveguide.pdf p6) runs the same 2.254 cm guide at 1 GHz and reaches the same verdict with  $f_c = 6.65$  GHz.

Problem 4: Ranking TM modes when  $a = 3b$  2+2

72 Ma

## QUESTION AS PRINTED 72 Ma [6+2+2]

Describe magic Tee based on S-parameters. Differentiate between dominant and degenerate modes. Consider a rectangular waveguide having dimension of  $a = 3b$  and find the dominant mode among TM<sub>01</sub>, TM<sub>10</sub>, TM<sub>11</sub>, TM<sub>21</sub>, TM<sub>12</sub>, TM<sub>02</sub> and TM<sub>20</sub>.

Step 1: put the cut-off formula in terms of  $a$  only

- $f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$ , and  $b = a/3 \Rightarrow n/b = 3n/a$ :

$$f_c = \frac{c}{2a} \sqrt{m^2 + 9n^2}$$

- So each mode is ranked by the number  $\sqrt{m^2 + 9n^2}$  (in units of  $c/2a$ ).

**Step 2: test every listed mode**

| Mode             | $(m, n)$ | Exists? (TM needs $m, n \geq 1$ ) | $\sqrt{m^2 + 9n^2}$ | Rank               |
|------------------|----------|-----------------------------------|---------------------|--------------------|
| TM <sub>01</sub> | (0, 1)   | no, $E_z \propto \sin 0 = 0$      | n/a                 | excluded           |
| TM <sub>10</sub> | (1, 0)   | no                                | n/a                 | excluded           |
| TM <sub>11</sub> | (1, 1)   | yes                               | $\sqrt{10} = 3.162$ | <b>1, dominant</b> |
| TM <sub>21</sub> | (2, 1)   | yes                               | $\sqrt{13} = 3.606$ | 2                  |
| TM <sub>12</sub> | (1, 2)   | yes                               | $\sqrt{37} = 6.083$ | 3                  |
| TM <sub>02</sub> | (0, 2)   | no                                | n/a                 | excluded           |
| TM <sub>20</sub> | (2, 0)   | no                                | n/a                 | excluded           |

- **Answer:** among the listed modes the dominant one is TM<sub>11</sub>,  $f_c = 3.162 c/2a$ .
- Worth one line: the guide's overall dominant mode is TE<sub>10</sub> at  $1 \times c/2a$ , which the list does not include.
- Riders: magic tee §4.5; dominant vs degenerate §4.2.

**Problem 5: Dominant TE and TM modes of a circular guide, with numbers** 2

75 Bh · plus the local worked example as practice

QUESTION AS PRINTED 75 Bh [8+2]

Derive the expression for the field strength for TM waves for a air-filled circular waveguide. Check the dominant mode in TE and TM modes.

**Step 1: cut-off in terms of the roots**

- TE<sub>nm</sub>:  $f_c = \frac{p'_{nm} c}{2\pi a}$ ; TM<sub>nm</sub>:  $f_c = \frac{p_{nm} c}{2\pi a}$ .
- Same  $a$  and  $c$  for every mode, so **the smallest root is the dominant mode**.

**Step 2: compare the smallest roots**

| Family | Smallest root                             | Mode             | Verdict                 |
|--------|-------------------------------------------|------------------|-------------------------|
| TE     | $p'_{11} = 1.841$ (first zero of $J'_1$ ) | TE <sub>11</sub> | <b>dominant TE mode</b> |
| TE     | $p'_{01} = 3.832$ (first zero of $J'_0$ ) | TE <sub>01</sub> | higher                  |
| TM     | $p_{01} = 2.405$ (first zero of $J_0$ )   | TM <sub>01</sub> | <b>dominant TM mode</b> |
| TM     | $p_{11} = 3.832$ (first zero of $J_1$ )   | TM <sub>11</sub> | higher                  |

- $1.841 < 2.405 \Rightarrow$  **TE<sub>11</sub> is the dominant mode of the guide**, TM<sub>01</sub> the dominant TM mode. The derivation part is theory §4.3.

**Practice (Waveguide.pdf p13): air-filled, radius 2 cm**

| Mode             | Substitution                                       | $f_c$           | $\lambda_c = 2\pi a/p$ |
|------------------|----------------------------------------------------|-----------------|------------------------|
| TE <sub>11</sub> | $1.8412 \times 3 \times 10^8 / (2\pi \times 0.02)$ | <b>4.40 GHz</b> | 6.83 cm                |
| TM <sub>01</sub> | $2.4048 \times 3 \times 10^8 / (2\pi \times 0.02)$ | <b>5.74 GHz</b> | 5.23 cm                |

- Does 3 GHz propagate?  $3 < 4.40 \text{ GHz} \Rightarrow$  **no**, not even in the dominant mode.

**Problem 6: Two magic tees joined at their E-arms** 10

77 Ch · derived by connecting two 4-port matrices; asserted in wg.py

QUESTION AS PRINTED 77 Ch [10]

Explain the properties of two Magic Tees if one connects their E-arms and derive its S-matrix.

**Given: the two tees**

- Each tee, ports (1, 2) collinear, 3 = E-arm, 4 = H-arm,  $r = 1/\sqrt{2}$  (§3.6):  $[S_T] = r \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$ .
- Tee A ports 1, 2, 3, 4; tee B ports 1', 2', 3', 4'. The E-arms 3 and 3' are joined, so they disappear as external ports.

- Joining means: the wave **leaving** arm 3 **enters** arm 3', and vice versa:  $a_{3'} = b_3, a_3 = b_{3'}$ .
- External ports left: 1, 2, 4 (tee A) and 1', 2', 4' (tee B)  $\Rightarrow$  a **6-port**,  $6 \times 6$  matrix.

### Step 1: drive port 1 with a unit wave ( $a_1 = 1$ , all other inputs zero)

- Inside tee A:  $b_3 = S_{31} = r, \quad b_4 = S_{41} = r, \quad b_2 = S_{21} = 0$ .
- $b_3 = r$  enters tee B's E-arm:  $a_{3'} = r$ .
- Inside tee B:  $b_{1'} = S_{13} a_{3'} = r \cdot r = \frac{1}{2}, \quad b_{2'} = S_{23} a_{3'} = -r \cdot r = -\frac{1}{2}, \quad b_{4'} = S_{43} a_{3'} = 0$ .
- Nothing reflects back into arm 3, because  $S_{3'3'} = 0$  (matched E-arm).
- Column for port 1:  $(b_1, b_2, b_4, b_{1'}, b_{2'}, b_{4'}) = (0, 0, r, \frac{1}{2}, -\frac{1}{2}, 0)$ .
- Power check:  $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1 \checkmark$ .

### Step 2: the other drives, same way

- Port 2:  $b_3 = S_{32} = -r, b_4 = r$ ; tee B receives  $-r \Rightarrow b_{1'} = -\frac{1}{2}, b_{2'} = +\frac{1}{2}$ .
- Port 4 (H-arm A):  $b_1 = b_2 = r$ , and  $b_3 = S_{34} = 0 \Rightarrow$  **nothing crosses** to tee B.
- Ports 1', 2', 4': mirror images, by symmetry of the connection.

### Step 3: the 6-port S-matrix

$$[S] = \begin{matrix} & \begin{matrix} 1 & 2 & 4 & 1' & 2' & 4' \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 4 \\ 1' \\ 2' \\ 4' \end{matrix} & \begin{bmatrix} 0 & 0 & r & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & r & -\frac{1}{2} & \frac{1}{2} & 0 \\ r & r & 0 & 0 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & r \\ -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & r \\ 0 & 0 & 0 & r & r & 0 \end{bmatrix} \end{matrix}$$

$r = 1/\sqrt{2}$ . Identical to the `connect()` result in `wg.py`.

### Step 4: properties

- **Matched**: all six diagonal entries are 0.
- **Reciprocal**:  $[S] = [S]^T$ .
- **Lossless**: every column's powers sum to 1.
- **Collinear isolation kept**:  $S_{12} = S_{1'2'} = 0$ .
- **H-arms isolated** from each other and from the far tee:  $S_{44'} = 0$ .
- **Difference channel crosses**: an antiphase pair at (1, 2) appears entirely at (1', 2'), again antiphase.
- **Sum channel stays**: an in-phase pair at (1, 2) leaves entirely by 4.
- Use: a sum/difference separator, e.g. two monopulse channels feeding a common difference path.

## Problem 7: Hybrid tee with a terminated arm 4+4+2

82 Ba · 70 Ma's "hybrid tee" [8] is answered by §4.5 plus this

QUESTION AS PRINTED 82 Ba [4+4+2]

Using proper S-Matrices, mention the properties of a Hybrid-Tee in the situation of (a) terminated E-arm, (b) terminated H-arm and (c) terminated co-planar arms.

### Step 0: what "terminated" does to a matrix

- Terminated = a **matched load**,  $\Gamma = 0$ : whatever leaves that arm is absorbed, nothing comes back, so  $a_k = 0$ .
- Therefore row  $k$  and column  $k$  simply **drop out**; the rest of the matrix is unchanged.
- Start from the magic tee, ports 1, 2 collinear (the co-planar arms), 3 = E, 4 = H.
- If the paper meant a **short**,  $\Gamma = -1$ , the answer is different; that is the 82 Bh case, §3.6.

| Case                                             | Reduced S-matrix                                                      | Properties                                                                                                                                                                                                                                                                        |
|--------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) E-arm terminated, ports (1, 2, 4)            | $\begin{bmatrix} 0 & 0 & r \\ 0 & 0 & r \\ r & r & 0 \end{bmatrix}$   | Matched and reciprocal. From the H-arm: <b>equal, in-phase</b> halves at 1 and 2, no loss. Ports 1 and 2 <b>isolated</b> . From port 1: only half reaches port 4, the other half is absorbed in the E-arm load, so the 3-port is <b>not lossless</b> . An in-phase power divider. |
| (b) H-arm terminated, ports (1, 2, 3)            | $\begin{bmatrix} 0 & 0 & r \\ 0 & 0 & -r \\ r & -r & 0 \end{bmatrix}$ | Matched and reciprocal. From the E-arm: <b>equal, antiphase</b> halves at 1 and 2. Ports 1 and 2 isolated. From port 1: half to port 3, half lost in the H-arm load. An antiphase (180°) power divider.                                                                           |
| (c) Both co-planar arms terminated, ports (3, 4) | $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$                        | E and H arms both <b>matched</b> and <b>completely isolated</b> . Any power entering either side arm splits into the two loads and nothing comes out anywhere: the tee acts as a matched termination for both.                                                                    |

- Why (a) and (b) **must** be lossy: they are matched, reciprocal 3-ports, and §3.5 proves no such 3-port can be lossless. The load is where the missing half goes.
- Column check, (a): column 1 =  $0 + 0 + \frac{1}{2} = \frac{1}{2}$  (half absorbed); column 4 =  $\frac{1}{2} + \frac{1}{2} = 1$  (lossless from the H-arm).

## Problem 8: Two magic tees connected in the H-plane 5

78 Ch · a short note; the Problem 6 method with the H-arms joined

### QUESTION AS PRINTED 78 Ch [5]

Write short notes on: Combinational effect of two magic fees connected in H-plane.

The paper's typos ("Combinational", "magic fees") are reproduced as printed; read "combinational effect of two magic tees".

### Step 1: join H-arm 4 to H-arm 4' ( $a_{4'} = b_4, a_4 = b_{4'}$ ) and follow port 1

- Tee A:  $b_3 = S_{31} = r, b_4 = S_{41} = r$ .
- $b_4 = r$  enters tee B's H-arm:  $b_{1'} = S_{14} r = \frac{1}{2}, b_{2'} = S_{24} r = +\frac{1}{2}$  (in phase),  $b_{3'} = S_{34} r = 0$ .
- From port 2:  $b_4 = S_{42} = +r$  as well, so tee B again gets  $+\frac{1}{2}, +\frac{1}{2}$ .

### Step 2: the 6-port S-matrix

$$[S] = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 1' & 2' & 3' \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 1' \\ 2' \\ 3' \end{matrix} & \begin{bmatrix} 0 & 0 & r & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -r & \frac{1}{2} & \frac{1}{2} & 0 \\ r & -r & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & r \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & -r \\ 0 & 0 & 0 & r & -r & 0 \end{bmatrix} \end{matrix}$$

### Short-note points

- Matched at all six ports, reciprocal, lossless.
- The **sum** (in-phase) signal crosses between the tees through the joined H-arms and re-appears **in phase** at 1', 2'.
- The **difference** signal stays in its own tee's E-arm.
- Collinear ports of each tee stay isolated; the two E-arms are isolated.
- The exact mirror of Problem 6: E-joining passes the difference, H-joining passes the sum.

## Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short   | Means                                | Short    | Means                                   |
|---------|--------------------------------------|----------|-----------------------------------------|
| RF      | radio frequency                      | MW, M/W  | microwave                               |
| EM, EMR | electromagnetic (radiation)          | TEM      | transverse electromagnetic mode         |
| TE      | transverse electric mode             | TM       | transverse magnetic mode                |
| WG      | waveguide                            | TL       | transmission line                       |
| VSWR    | voltage standing wave ratio          | SWR      | standing wave ratio                     |
| VNA     | vector network analyzer              | SA       | spectrum analyzer                       |
| DUT     | device under test                    | S-params | scattering parameters                   |
| RL      | return loss                          | IL       | insertion loss                          |
| ML      | mismatch loss                        | NF       | noise figure                            |
| BW      | bandwidth                            | SNR      | signal to noise ratio                   |
| Tx, TX  | transmitter                          | Rx, RX   | receiver                                |
| BTS     | base transceiver station             | GSM      | global system for mobile communications |
| EIRP    | effective isotropic radiated power   | ERP      | effective radiated power                |
| SAR     | specific absorption rate             | SARPs    | standards and recommended practices     |
| HERP    | hazards of EM radiation to personnel | RADAR    | radio detection and ranging             |
| ASR     | airport surveillance radar           | CW       | continuous wave                         |

  

| Short     | Means                                                       | Short   | Means                                                   |
|-----------|-------------------------------------------------------------|---------|---------------------------------------------------------|
| TWT, TWTA | traveling wave tube (amplifier)                             | BWO     | backward wave oscillator                                |
| CFA       | cross-field amplifier                                       | LNA     | low noise amplifier                                     |
| HPA       | high power amplifier                                        | SWS     | slow wave structure                                     |
| O-type    | linear beam tube (axial $B$ -field)                         | M-type  | crossed-field tube ( $E \perp B$ )                      |
| MASER     | microwave amplification by stimulated emission of radiation | LASER   | light amplification by stimulated emission of radiation |
| MESFET    | metal-semiconductor FET                                     | HEMT    | high electron mobility transistor                       |
| GaAs      | gallium arsenide                                            | FET     | field effect transistor                                 |
| BJT       | bipolar junction transistor                                 | PIN     | p-type, intrinsic, n-type diode                         |
| IMPATT    | impact ionization avalanche transit-time                    | TRAPATT | trapped plasma avalanche triggered transit              |
| LPF       | low pass filter                                             | HPF     | high pass filter                                        |
| BPF       | band pass filter                                            | BSF     | band stop filter                                        |
| MAG       | maximum available gain                                      | MSG     | maximum stable gain                                     |
| MIC       | microwave integrated circuit                                | MMIC    | monolithic microwave integrated circuit                 |
| ADS       | advanced design system                                      | IF      | intermediate frequency                                  |
| LO        | local oscillator                                            | RFIC    | radio frequency integrated circuit                      |
| PCB       | printed circuit board                                       | SMA     | subminiature version A connector                        |

  

| Short    | Means                | Short  | Means                            |
|----------|----------------------|--------|----------------------------------|
| attr(s)  | attribute(s)         | no.    | number                           |
| freq     | frequency            | info   | information                      |
| ops      | operations           | config | configuration                    |
| rm       | remove               | vs, Vs | versus                           |
| w/, w/o  | with, without        | Eg     | example / for example            |
| +Fig     | draw a figure        | (I)    | importance / need / significance |
| →        | leads to / therefore | +      | and                              |
| =        | is / equals          | norm   | normalized                       |
| max, min | maximum, minimum     | calc   | calculation / calculate          |
| eq, eqn  | equation             | approx | approximately                    |
| pts      | points               | coeff  | coefficient                      |
| temp     | temperature          | params | parameters                       |

**Standard and component terms** (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ( $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ ) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

**Key mathematical symbols:**  $Z_0$  characteristic impedance ·  $Y_0$  characteristic admittance ·  $\Gamma$  reflection coefficient ·  $S$  scattering matrix ·  $K$  Rollett's stability factor ·  $\Delta$  determinant of S-matrix ·  $\mu$  single parameter stability metric ·  $G_T$  transducer gain ·  $G_{TU}$  unilateral transducer gain ·  $f_c$  cutoff frequency ·  $\lambda_g$  guide wavelength ·  $v_p$  phase velocity ·  $v_g$  group velocity.

**Exam-paper month codes** (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

**Bold** year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

**Tier chips** over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth  $\geq 6$  marks.

**Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.